NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

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Introduction to

Internet of Things

Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

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QUESTION 1:

What are some of the other names of the smart grids?

a. Energy internet
b. Electricity with a hand
c. Electrolyte
d. All of these

Correct Answer: a. Energy internet

Detailed Solution: Smart grids are also known as Electricity with a brain, Energy internet, and Electronet. (Please refer Lecture 51@8:33)

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QUESTION 2:

Smart Grid follows which type of flow of energy?

a. Bidirectional

b. Unidirectional

c. Both a and b

d. Neither a nor b

Correct Answer: a. Bidirectional

Detailed Solution: Smart grid follows bidirectional flow of energy. (Please refer Lecture 51@5:41)

QUESTION 3:

Which of the following is a benefit associated with smart grids for customers?

a. Reduced transmission of electricity
b. Higher electricity rates
c. Less security
d. Different pricing options

Correct Answer: d. Different pricing options

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Detailed Solution: Smart Grids provide different pricing options for customers, lower electricity rates and Improved security. (Please refer Lecture 51@16:33)

QUESTION 4:

Which of the following is a property of smart grid related to power system efficiency?

a. Power monitoring
b. Asset Management
c. Distribution automation
d. All of these

Correct Answer: d. All of these

Detailed Solution: Power monitoring, Asset Management and Distribution automation are all the properties of smart grid related to power system efficiency. (Please refer Lecture 51 @22:21)

QUESTION 5:

Fill in the blank.

In _____, a home can have power from distributed resources.

a. Peaking
b. Islanding
c. Neither a nor b
d. Both a and b

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Correct Answer: b. Islanding

Detailed Solution: In islanding, a home can have power from distributed resources.(Please refer Lecture 51 @31:20)

QUESTION 6:

Which of the following aggregate the energy consumption or energy request of a certain geographical area?

a. DAUs
b. Meter Data Management System
c. Buffer
d. None of these

Correct Answer: a. DAUs

Detailed Solution: DAUs aggregate the energy consumption or energy request of a certain geographical area. (Please refer Lecture 52@15:17)

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QUESTION 7:

Which of the following statements are true about Dynamic System Attacks?

Statement I: Replay attacks compromises sensors and monitors their outputs.

Statement II: D-DIA can lead to a system collapse.

Statement III: Covert attack is a closed loop version of replay attacks.

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statement I, II and III

Correct Answer: d. Statements I, II and III

Detailed Solution: Replay attacks compromises sensors and monitors their outputs. D-DIA can lead to a system collapse. Covert attack is a closed loop version of replay attacks. (Please refer Lecture 52@20:03)

QUESTION 8:

Fill in the blank.

_____ is performed by compromising the availability of the grid component.

a. Gateway
b. Data injection attacks
c. AMH
d. Denial of service

Correct Answer: d. Denial of service

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Detailed Solution: Denial of service is performed by compromising the availability of the grid component. (Please refer Lecture 52@19:52)

QUESTION 9:

Fill in the blank.

IIoT is a network of _____ .

a. Physical objects
b. Systems
c. Platforms
d. All of these

Correct Answer: d. All of these

Detailed Solution: IIoT is a network of Physical objects, Systems and Platforms. (Please refer Lecture 52@14:46)

QUESTION 10:

Fill in the blank. _____ is based on a wrap and re-use approach.

a. AoT

b. PoT

c. ToII

d. IIoT

Correct Answer: d. IIoT

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Detailed Solution: IIoT is based on the wrap and re-use approach. (Please refer Lecture 53@9:51)

QUESTION 11:

Mechanized production is a part of the 4th Industrial Revolution.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Mechanized production is a part of the 1st Industrial Revolution. (Please refer Lecture 53@10:41)

QUESTION 12:

Which of the following happened in the 3rd Revolution?

a. Mass production
b. IIoT
c. Mechanized production
d. None of these

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Correct Answer: d. None of these

Detailed Solution: Internet evolution and automation happened in the 3rd Revolution.(Please refer Lecture 53@10:06)

QUESTION 13:

Which of the following is used to improve the interoperability of different systems?

a. Standardization
b. Optimization
c. Monitoring
d. None of these

Correct Answer: a. Standardization

Detailed Solution: Standardization improves the interoperability of different systems. (Please refer Lecture 54@16:09)

QUESTION 14:

Big data is characterized by 7 Vs.

a. True
b. False

Correct Answer: a. True

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Detailed Solution: Big data is characterized by 7 Vs. (Please refer Lecture 55@10:36)

QUESTION 15:

What is Hadoop used in?

a. Distributed processing of large datasets
b. Small clusters of computers
c. Both a and b
d. Neither a nor b

Correct Answer: a. Distributed processing of large datasets

Detailed Solution: Hadoop is used for distributed processing of large datasets across Large clusters of computers (Please refer Lecture 55@23:18)

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